

RELATORIO TAF x METAR - grupo NavBrasil

Gerado 2026-09-01 11:15 | 43 aerodromos | fonte: excel/malha.csv + tabelas metars / tafs

Secoes: 1) presenca por estacao x periodo 2) fenomenos perigosos, acuracia 3) perigosos por estacao x periodo 4) consolidado. Matrices por aeroporto em output/*_NavBrasil.csv.

TAF REDEMET vs METAR - Estacao x Periodo | grupo NavBrasil

Gerado 2026-09-01 11:15 | malha.csv | 129,443 voos comparaveis | 24 aeroportos com dado

Regra: acerto = TAF e METAR concordam na presenca de fenomeno (ambos com condicao, ou ambos NSW). Falso positivo = TAF previu, METAR nao confirmou. Miss = METAR teve condicao, TAF nao previu.
Janela: 'Hora Chegada Alternado' UTC. Estacoes astronomicas (hem. sul).

Geral

Comparaveis	Certo	%	Falso Positivo	Miss
129,443	110,993	85.7%	10,458	7,992

Por estacao

estacao	Comparaveis	Certo	%	Falso Pos.	Miss
Verao	31,817	25,422	79.9%	3,513	2,882
Outono	32,219	27,317	84.8%	2,931	1,971
Inverno	33,582	30,249	90.1%	1,850	1,483
Primavera	31,825	28,005	88.0%	2,164	1,656

Por periodo (UTC)

(UTC)	Comparaveis	Certo	%	Falso Pos.	Miss
00-06	29,116	25,318	87.0%	1,874	1,924
06-12	17,895	14,792	82.7%	1,880	1,223
12-18	39,387	34,975	88.8%	1,847	2,565
18-24	43,045	35,908	83.4%	4,857	2,280

Estacao x periodo (% acerto / n)

Estacao \ Periodo	00-06	06-12	12-18	18-24	TODOS
Verao	82.0% (7,272)	80.6% (4,427)	85.1% (9,694)	73.3% (10,424)	79.9% (31,817)
Outono	85.9% (7,006)	78.8% (4,243)	87.2% (9,876)	84.2% (11,094)	84.8% (32,219)
Inverno	90.2% (7,609)	83.7% (4,719)	92.3% (10,178)	90.6% (11,076)	90.1% (33,582)
Primavera	89.5% (7,229)	87.3% (4,506)	90.4% (9,639)	85.1% (10,451)	88.0% (31,825)
TODAS	87.0% (29,116)	82.7% (17,895)	88.8% (39,387)	83.4% (43,045)	85.7% (129,443)

Aeroportos | grupo NavBrasil

Gerado 2026-09-01 11:15 | 24 aeroportos com voo comparavel | ordenado por volume

ICAO	Comparaveis	Certo	%	Falso Pos.	Miss
SBKP	26,616	23,184	87.1%	1,475	1,957
SBJP	25,587	21,969	85.9%	2,512	1,106
SBRP	24,578	21,924	89.2%	1,576	1,078
SBMQ	7,910	6,025	76.2%	1,153	732
SBGO	7,726	6,584	85.2%	532	610
SBNF	7,598	6,228	82.0%	821	549
SBGR	6,249	5,403	86.5%	330	516
SBAR	5,782	4,799	83.0%	451	532
SBUR	4,174	3,653	87.5%	259	262
SBTE	2,169	1,895	87.4%	191	83
SBCJ	2,080	1,703	81.9%	263	114
SBLO	1,876	1,661	88.5%	110	105
SBJV	1,490	1,185	79.5%	167	138
SBMA	1,077	910	84.5%	128	39
SBAE	789	708	89.7%	56	25
SBKG	738	588	79.7%	123	27
SBHT	669	539	80.6%	109	21
SBMK	502	460	91.6%	24	18
SBUL	437	396	90.6%	19	22
SBPL	359	342	95.3%	11	6
SBSN	337	279	82.8%	33	25
SBIZ	327	280	85.6%	39	8
SBTF	256	181	70.7%	64	11
SBVT	117	97	82.9%	12	8

Acuracia - fenomenos perigosos | grupo NavBrasil

Gerado 2026-09-01 11:15 | malha.csv | 129,443 voos comparaveis | TAF REDEMET vs METAR observado

Lista: FZDZ FZRA FZFG TS +TS +RA +SHRA +SN +SHSN TSRA. 'QUALQUER perigosa' = o voo tem pelo menos um desses tokens. Acuracia = acerto / (acerto + falso positivo + miss).

Leitura	Voos c/ perigo	Acerto	Falso pos.	Miss	Acuracia	Deteccao	Precisao
exato	7,160	318	6,028	814	4.4%	28.1%	5.0%
sem intensidade	7,811	554	5,812	1,445	7.1%	27.7%	8.7%

Deteccao = acerto / observados (1 - miss). **Precisao** = acerto / sinalizados (1 - falso positivo).

Por fenomeno

Gerado 2026-09-01 11:15 | 129,443 voos comparaveis

Visao EXATA

Fenomeno	TAF previu	Confirmado	Precisao %	METAR teve	Detectado %	Falso pos.	Miss
TS	1,275	35	2.7%	612	5.7%	1,240	577
+RA	1	0	0.0%	91	0.0%	1	91
+SHRA	0	0	-	8	0.0%	0	8
+TSRA	1	0	0.0%	138	0.0%	1	138
TSRA	5,069	58	1.1%	283	20.5%	5,011	225

Visao SEM INTENSIDADE

Fenomeno	TAF previu	Confirmado	Precisao %	METAR teve	Detectado %	Falso pos.	Miss
FZDZ	0	0	-	0	-	0	0
FZFG	0	0	-	0	-	0	0
FZRA	0	0	-	0	-	0	0
TS	1,275	35	2.7%	612	5.7%	1,240	577
TSRA	5,090	293	5.8%	1,288	22.7%	4,797	995
+RA	1	0	0.0%	91	0.0%	1	91
+SHRA	0	0	-	8	0.0%	0	8
+SHSN	0	0	-	0	-	0	0
+SN	0	0	-	0	-	0	0

Condicoes perigosas - TAF REDEMET vs METAR | grupo NavBrasil

Gerado 2026-09-01 11:15 | malha.csv | 129,443 voos comparaveis | 24 aeroportos

Acerto TAF % = dos voos em que o TAF previu a condicao, quantos o METAR confirmou (match exato).
Detectado % = dos voos com a condicao no METAR, quantos o TAF previra.
Congelantes/+TS/+SHRA/+SN/+SHSN omitidos (zero no TAF). Estacoes astronomicas (hem. sul), periodos UTC.

Visao EXATA (token identico) - geral

Cond	TAF previu	Confirmado	Acerto TAF %	METAR teve	Detectado %
TS	1,275	35	2.7%	612	5.7%
+RA	1	0	0.0%	91	0.0%
+TSRA	1	0	0.0%	138	0.0%
TSRA	5,069	58	1.1%	283	20.5%

Visao SEM INTENSIDADE (+/- removido) - geral

Cond	TAF previu	Confirmado	Acerto TAF %	METAR teve	Detectado %
TS	1,275	35	2.7%	612	5.7%
TSRA	5,090	293	5.8%	1,288	22.7%
+RA	1	0	0.0%	91	0.0%

Por estacao - visao exata

estacao \ cond	TS	+RA	+TSRA	TSRA
Verao	1.2% (575)	-	-	0.6% (2,190)
Outono	1.5% (341)	-	-	1.3% (1,132)
Inverno	4.3% (115)	-	0.0% (1)	1.3% (390)
Primavera	7.4% (244)	0.0% (1)	-	1.8% (1,357)

Por periodo (UTC) - visao exata

periodo \ cond	TS	+RA	+TSRA	TSRA
00-06	1.2% (85)	-	-	0.8% (397)
06-12	0.0% (23)	0.0% (1)	0.0% (1)	1.4% (207)
12-18	2.3% (86)	-	-	0.9% (586)
18-24	3.0% (1,081)	-	-	1.2% (3,879)

TSRA - estacao x periodo (acerto TAF % / n previsto)

Estacao \ Periodo	00-06	06-12	12-18	18-24	TODOS
Verao	0.7% (141)	1.3% (75)	0.0% (164)	0.7% (1,810)	0.6% (2,190)
Outono	0.0% (96)	0.0% (70)	0.7% (151)	1.7% (815)	1.3% (1,132)
Inverno	2.1% (47)	0.0% (28)	3.3% (91)	0.4% (224)	1.3% (390)
Primavera	0.9% (113)	5.9% (34)	0.6% (180)	1.9% (1,030)	1.8% (1,357)
TODAS	0.8% (397)	1.4% (207)	0.9% (586)	1.2% (3,879)	1.1% (5,069)

TS - estacao x periodo (acerto TAF % / n previsto)

Estacao \ Periodo	00-06	06-12	12-18	18-24	TODOS
Verao	0.0% (24)	0.0% (8)	5.0% (40)	1.0% (503)	1.2% (575)
Outono	0.0% (21)	0.0% (2)	0.0% (20)	1.7% (298)	1.5% (341)
Inverno	12.5% (8)	0.0% (7)	0.0% (3)	4.1% (97)	4.3% (115)
Primavera	0.0% (32)	0.0% (6)	0.0% (23)	9.8% (183)	7.4% (244)
TODAS	1.2% (85)	0.0% (23)	2.3% (86)	3.0% (1,081)	2.7% (1,275)

Aeroportos - TS/TSRA (visao exata) | grupo NavBrasil

Gerado 2026-09-01 11:15 | 24 aeroportos com TS/TSRA no TAF ou METAR. Ordenado por nº previsto.

ICAO	Voos	TAF previu	Confirmado	Acerto %	METAR teve	Detectado %
SBRP	24,578	1,304	71	5.4%	199	35.7%
SBKP	26,616	1,231	40	3.2%	199	20.1%
SBMQ	7,910	1,186	44	3.7%	112	39.3%
SBGO	7,726	705	25	3.5%	82	30.5%
SBUR	4,174	357	33	9.2%	71	46.5%
SBJP	25,587	275	9	3.3%	21	42.9%
SBNF	7,598	206	8	3.9%	66	12.1%
SBTE	2,169	195	5	2.6%	18	27.8%
SBGR	6,249	155	7	4.5%	33	21.2%
SBMA	1,077	138	2	1.4%	5	40.0%
SBLO	1,876	96	5	5.2%	15	33.3%
SBCJ	2,080	75	1	1.3%	32	3.1%
SBJV	1,490	64	5	7.8%	14	35.7%
SBHT	669	61	1	1.6%	6	16.7%
SBTF	256	58	0	0.0%	2	0.0%
SBAE	789	54	0	0.0%	0	-
SBAR	5,782	43	0	0.0%	7	0.0%
SBIZ	327	39	1	2.6%	4	25.0%
SBSN	337	30	0	0.0%	3	0.0%
SBUL	437	23	4	17.4%	5	80.0%
SBMK	502	19	0	0.0%	0	-
SBKG	738	18	0	0.0%	0	-
SBPL	359	9	1	11.1%	1	100.0%
SBVT	117	3	0	0.0%	0	-

TAF REDEMET vs METAR | grupo NavBrasil

Gerado 2026-09-01 11:15 | comparaveis: 129,443 de 131,086

Regra: acerto quando TAF e METAR concordam na presenca de fenomeno (ambos com condicao, ou ambos NSW).

	Certo	%	Falso Positivo	Miss
TAF REDEMET	110,993	85.7%	10,458	7,992

TAF REDEMET vs METAR - Condições Perigosas (match exato)

Gerado 2026-09-01 11:15 | comparáveis: 129,443

Condição	TAF previu	METAR igual	Errado	Acerto %
-FZDZ	0	0	0	0.0%
FZDZ	0	0	0	0.0%
+FZDZ	0	0	0	0.0%
-FZRA	0	0	0	0.0%
FZRA	0	0	0	0.0%
+FZRA	0	0	0	0.0%
FZFG	0	0	0	0.0%
TS	1,275	35	1,240	2.7%
+TS	0	0	0	0.0%
+RA	1	0	1	0.0%
+SHRA	0	0	0	0.0%
+SN	0	0	0	0.0%
+SHSN	0	0	0	0.0%
+TSRA	1	0	1	0.0%
TSRA	5,069	58	5,011	1.1%
TOTAL	6,346	93	6,253	1.5%

O que o METAR mostrou, quando o TAF previu cada condição:

TS (previsto 1,275x)

METAR real	Qtd	%
NSW	898	70.4%
VCSH	125	9.8%
-RA	93	7.3%
VCTS	34	2.7%
-TSRA	31	2.4%
TS	18	1.4%
TS VCSH	17	1.3%
+TSRA	12	0.9%
Other	47	3.7%

+RA (previsto 1x)

METAR real	Qtd	%
-RA BR	1	100.0%

+TSRA (previsto 1x)

METAR real	Qtd	%
-RA	1	100.0%

TSRA (previsto 5,069x)

METAR real	Qtd	%
NSW	3,292	64.9%
-RA	618	12.2%
VCSH	347	6.8%
-TSRA	168	3.3%
TS VCSH	93	1.8%
VCTS	71	1.4%
RA	67	1.3%
-RA BR	65	1.3%
Other	348	6.9%

TAF REDEMET vs METAR - Perigosas (intensidade ignorada)

Gerado 2026-09-01 11:15 | comparaveis: 129,443

Condicao base	TAF previu	METAR igual	Errado	Acerto %
FZDZ	0	0	0	0.0%
FZFG	0	0	0	0.0%
FZRA	0	0	0	0.0%
TS	1,275	35	1,240	2.7%
TSRA	5,090	293	4,797	5.8%
+RA	1	0	1	0.0%
+SHRA	0	0	0	0.0%
+SHSN	0	0	0	0.0%
+SN	0	0	0	0.0%
TOTAL	6,366	328	6,038	5.2%